

The analysis DNN and the VBF selector

$H \rightarrow \mu\mu$ Run 3 analysis (skim_v4): purpose, training, architecture, results and fit plan

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Abstract

The Run 3 $H \rightarrow \mu\mu$ analysis uses two neural networks with different jobs.

- The **analysis DNN** (`DNN_NNOutput`) separates the $H \rightarrow \mu\mu$ signal (VBF H + ggH) from the background inside the cut-based VBF category. Its distribution is the fit variable of the COMBINE likelihood.
- The **VBF selector** (`VBFSel_NNOutput`) tags the *VBF topology* (VBF H + electroweak Zjj) against everything else, in all events with at least two jets. It does not use the dimuon mass. Its first use is to measure the EWK Z normalisation in the dimuon-mass sidebands, which is the leading uncertainty of the DNN fit. It could later define an alternative VBF category.

This note covers, for both networks: the signal definition, training region and sample, input features, architecture and training procedure, deployment in the framework, and every result available on 7 October 2026. The last part lists all fits, with and without systematic uncertainties, with and without the VBF split by tag-jet pseudorapidity (CC/CF/FF). Each fit is marked done, invalid, in progress or missing, and the planned order is given.

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1 Analysis context

1.1 Event selection, categories and mass regions

Baseline selection.

- Two opposite-sign muons, HLT_IsoMu24 with trigger matching.
- $p_T(\mu_1) > 26$ GeV and $p_T(\mu_2) > 20$ GeV, $|\eta| < 2.4$.
- Medium ID, $|d_z| < 0.1$ cm, $|d_{xy}| < 0.05$ cm.
- Relative isolation with the FSR photon subtracted, below 0.25.
- Exactly two loose muons and no good electron.
- b-jet veto (UParT/PNet working points).
- The muon four-momentum includes the beam-spot constraint (bsc), scale and resolution corrections (ScaRe) and FSR recovery.

Jets.

- AK4, JEC and JER applied, tight ID, jet veto maps.
- Cleaned against the selected leptons.
- “Horn” veto against noisy forward jets:
 - 2022–23: every jet with $|\eta| \geq 2.5$ and $p_T < 50$ GeV is removed;
 - 2024–26: only jets with $2.5 \leq |\eta| < 3.0$ and $p_T < 50$ GeV are removed.

Categories.

- **VBF** (cut-based): a VBF jet pair with $p_T > 35/25$ GeV, $m_{jj} > 400$ GeV and $|\Delta\eta_{jj}| > 2.5$. The production pair is the one with the largest m_{jj} .
- **ggF**: baseline events that fail the VBF requirement. It has sub-categories by jet multiplicity (0J, 1J, 2J, ≥ 2 J).
- **baseline_ge2J**: baseline with at least two jets, which equals **ggF_ge2J** + **VBF_ge2J**.
- **CC/CF/FF**: the VBF category split by the $|\eta|$ of the two tag jets, C if $|\eta| < 2.5$ and F if $|\eta| \geq 2.5$. “incl” is the unsplit category.

region	$m_{\mu\mu}$ [GeV]	role
Z sideband (ZSB)	76–110	control region; EWK Z fit
H sideband (HSB)	110–115 and 135–150	control region in the $H \rightarrow \mu\mu$ fit; EWK Z fit
Signal_Fit (SR)	115–135	signal region (blinded above DNN = 0.8)
Signal_ext	110–150	plots

Table 1: Dimuon-mass regions.

Eras and luminosity. 2022, 2022EE, 2023, 2023BPix, 2024, 2025 and 2026: 308.8 fb^{−1} in total, 283.0 fb^{−1} for 2022–2025.

Plots. All plots quoted here are under https://vdamante.web.cern.ch/H_mumu/skim_v4/, which is /eos/user/v/vdamante/www/H_mumu/skim_v4/.

1.2 Elements shared by the two networks

- **Network type.** Both are fully connected binary classifiers (MLP) with a sigmoid output.
- **k-fold.** Both use 4 folds.
 - Model k is trained without the events with `FullEventId % 4 = k`.
 - In the framework, each event is scored by model `FullEventId % 4`, so no event is ever scored by a model that saw it in training.
- **ONNX.** The models are exported to ONNX with the input standardisation $(x - \mu)/\sigma$ inside the graph. Inference uses `onnxruntime` on CPU (`common/dnn_application.py`), and the histogram maker books the score for every variable ending in `_NNOutput`.
- **Systematic variations.**
 - For the jet (JES, JER) and muon (scale/resolution) variations the score is *re-evaluated* on the shifted inputs (`ApplyDNNVariations`), together with the derived observables (Zepfenfeld variable, R_{pT} , soft activity, ...).
 - Before 29 September 2026 this was not the case: the score fell back to the nominal value, so jet and muon systematics only moved events across the selection. JES Absolute changes the DNN score of 16% of the events and the VBF selector score of 52%.

2 The analysis DNN (DNN_NNOutput)

2.1 Purpose and signal definition

The DNN classifies events of the cut-based VBF category in the signal mass window.

- **Signal:** VBF H (`VBFHto2Mu`) and ggH (`GluGluHto2Mu`), both with $m_H = 125$ GeV. The ggH events that pass the VBF selection are treated as signal, and the fit has a single signal strength r for the sum.
- **Background:** everything else.
 - Training: Drell–Yan ($m_{\ell\ell}$ 105–160 GeV), electroweak Zjj ($m_{\ell\ell}$ 105–160 GeV, Herwig), $t\bar{t}$, VV.
 - Evaluation and fit: also single top, tW , $t\bar{t}X$, VVV, W+jets, VH and $t\bar{t}H$.
- **Fit:** the score, binned in 14 bins per era, is fitted in the SR and in the HSB of the VBF category. The HSB constrains the background normalisations and shapes.

2.2 Payloads and provenance

payload	directories	status
updated_DNN (production)	<code>common/updated_DNN_configs</code> , <code>common/updated_DNN_models</code>	used in every era (default <code>model_set="updated"</code>)
legacy	<code>common/dnn_configs</code> , <code>common/dnn_models</code> (_2024 for 2024–26)	<code>model_set="legacy"</code> , comparisons only
VBFNet (PR #21)	<code>common/vbfnet_configs</code> , <code>common/vbfnet_models</code>	not usable: saturates, AUC 0.39–0.47 on skim_v4

Table 2: DNN payloads known to `common/dnn_application.py`.

The production model was not trained in this repository. The config field `results_dir` points to `/eos/user/a/ayeagle/H_mumu/results/OGDNN_big_run/`. The description below is reconstructed

from three sources: the deployed config, the ONNX graph, and the training-tuple configuration kept in the repository (`studies/dnn/traintuple_config_DNN.yaml`).

2.3 Training region and sample

selection	(Signal_Fit) & (VBF): cut-based VBF category, $115 < m_{\mu\mu} < 135$ GeV
processes	DY $m_{\ell\ell}$ 105–160, $t\bar{t}$, EWK Zjj 105–160 (Herwig), VV, VBF H and ggH (amc@NLO)
eras	2022–2025, distinguished by the input <code>era_code</code> (0 to 5); 2026 is evaluated with the 2025 code
labels	signal = VBFHto2Mu, GluGluHto2Mu; everything else is background
weights	<code>weight__Central</code> in absolute value (<code>abs_train_weight=true</code>); signal and background sums equalised (<code>equalize_for_training</code>)
inputs	standardised with the training mean and standard deviation (<code>renorm_inputs</code>)
folds	$k = 4$; 33% of the training folds used for validation

Table 3: Training set-up of the production DNN.

2.4 Input features (32)

feature	meaning
<code>era_code</code>	era index, 0 = 2022 ... 5 = 2025 (2026 $\rightarrow$ 5)
<code>mu1_pt</code> , <code>mu1_eta</code> , <code>mu2_pt</code> , <code>mu2_eta</code>	leading and subleading muon kinematics
<code>m_mumu</code> , <code>pt_mumu</code> , <code>eta_mumu</code> , <code>phi_mumu</code> , <code>y_mumu</code>	dimuon system
<code>dR_mumu</code>	ΔR between the muons
<code>cosTheta_CS</code> , <code>phi_CS</code>	Collins–Soper angles
<code>R_pt</code>	p_T balance of the dimuon + tag-jet system
<code>minDeltaEtaSigned</code> , <code>minDeltaPhi</code>	smallest $\Delta\eta$ and $\Delta\phi$ between the dimuon and a tag jet
<code>Zeppenfeld_Var</code>	dimuon rapidity relative to the tag jets, $(y_{\mu\mu} - \bar{y}_{jj})/ \Delta y_{jj} $
<code>pt_centrality</code>	p_T centrality of the dimuon with respect to the tag jets
<code>vbffjet{1,2}_pt</code> , <code>_eta</code> , <code>_y</code>	tag-jet kinematics
<code>vbffjet{1,2}_btagPNetQvG</code>	quark/gluon discriminant of the tag jets
<code>m_jj</code> , <code>delta_eta_jj</code> , <code>pt_vbffj1j2</code>	dijet mass, $ \Delta\eta $ and p_T of the tag-jet pair
<code>N_SelectedJets</code>	jet multiplicity
<code>SoftJetCleanedActivity_N</code> , <code>_ptSum</code>	soft hadronic activity outside the tag jets and muons

Table 4: Inputs of the production DNN, in the payload order (32 features).

$m_{\mu\mu}$ is an input, so the score can sculpt the mass distribution. The fit uses the score in fixed mass regions, so this does not bias the fit, but it ties the score shape to the mass resolution. The retraining (§2.13) has a second candidate without `m_mumu`.

2.5 Architecture and training procedure

layers (from the ONNX graph)	$32 \rightarrow 86 \rightarrow 86 \rightarrow 86 \rightarrow 86 \rightarrow 1$ (Gemm)
activation	ReLU; sigmoid output
regularisation	none: <code>dropout=0</code> , <code>batch_renorm=false</code>
loss	weighted binary cross-entropy
optimiser	Adam, learning rate 10^{-3} , weight decay 0
batch size	10 000
epochs	<code>epochs=1</code> in the deployed config, early stopping with patience 10
trainable parameters	$32 \cdot 86 + 3 \cdot 86^2 + 86 + 5$ biases $\approx 2.5 \times 10^4$

Table 5: Production DNN (`common/updated_DNN_configs/config.toml`).

The deployed config sets one epoch. The retraining config of 30 September sets 200 epochs with early stopping, to correct the training length.

Legacy payload.

- 26 inputs: the same as above without `era_code`, the tag-jet rapidities, `pt_vbfj1j2`, `N_SelectedJets` and the soft activity, and with `pt_jj`.
- ONNX graph $26 \rightarrow 50 \rightarrow 50 \rightarrow 50 \rightarrow 50 \rightarrow 1$. The config says [52,26,26,13]; the graph is authoritative.
- Trained on `VBF_JetVeto & Signal_Fit`, with Adam (lr 10^{-4}), batch 1200, 64 epochs and patience 20.

2.6 Deployment and templates

- **Score.** `DNN_NNOutput` is computed in `histograms/hist_maker.py` for the nominal and for every jet/muon variation.
- **Fit binning.** 14 bins per era (`config/plot/histograms.yaml`); the SR is split at `DNN = 0.8` for blinding.
- **Histogram campaigns:**
 - `dnn_vbf`: VBF category, five mass regions, all systematic families, DY/EWK jet components. Used for the plots.
 - `dnn_vbf_eta_regions_syst`: the same with the CC/CF/FF split; the `incl` sub-directory is the unsplit category. These are *the fit templates from now on*, both inclusive and split.
 - `dnn_vbf_eta_regions`: CC/CF/FF, Central only. It was used for the categorisation study.
 - `dnn_vbf_weighted` (17 September): produced before the score was re-evaluated for jet and muon variations, so its JES/JER/muon-scale templates are wrong. Every DNN fit with systematics quoted below used it.
- **DY reweighting.** DY carries three reweightings: jet component, $p_T(\ell\ell)$ and N_{jets} . DY is split into *Hard* and *PU1/PU2* components, according to whether the tag jets match generator jets or come from pile-up.

2.7 Results: separation power

era	2022	2022EE	2023	2023BPix	2024	2025	2026*
weighted AUC	0.905	0.903	0.908	0.899	0.890	0.888	0.905
S (events)	1.8	6.0	4.0	2.2	43.3	45.2	—
B (events)	382	1153	923	498	9267	9323	—
best cut	0.921	0.919	0.920	0.950	0.944	0.871	—
best $S/\sqrt{S+B}$	0.287	0.491	0.375	0.275	1.113	1.086	—
signal eff. at best cut	0.44	0.44	0.45	0.30	0.24	0.40	—

Table 6: DNN_NNOutput in Signal_Fit_VBF, from the histograms with the DY weights (results/skim_v4/dnn_performance/dy_weights_weighted/). *2026: ROC from the skims, 30 files per dataset (roc_from_skims). The DY weights change the AUC by -0.005 (2024) and the best $S/\sqrt{S+B}$ by $+2.5\%$.

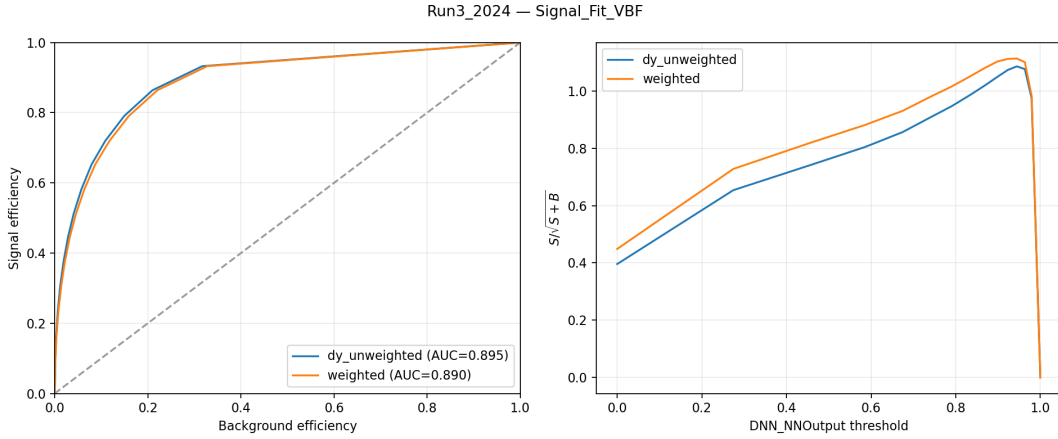


Figure 1: DNN, 2024, SR of the VBF category: ROC curve and $S/\sqrt{S+B}$ as a function of the threshold, with and without the DY weights. [All eras.](#)

2.8 Results: binning of the score

studies/dnn/optimize_dnn_binning.py scans contiguous partitions that maximise the binned Asimov significance. It uses statistics only and requires at least 10 background events per bin.

	2022	2022EE	2023	2023BPix	2024	2025
best single cut	0.276	0.507	0.385	0.267	1.171	1.114
5 bins	0.292	0.575	0.438	0.289	1.540	1.493
gain	+6%	+13.5%	+13.7%	+8.4%	+31.5%	+34%

Table 7: Binned Asimov significance, statistics only.

Five bins recover 98–99.8% of the maximum, and the last bin sits on the 10-event floor, so the MC statistics sets the edges. The fit uses 14 bins; autoMCStats costs about 0.08σ . 2026 is not in this scan.

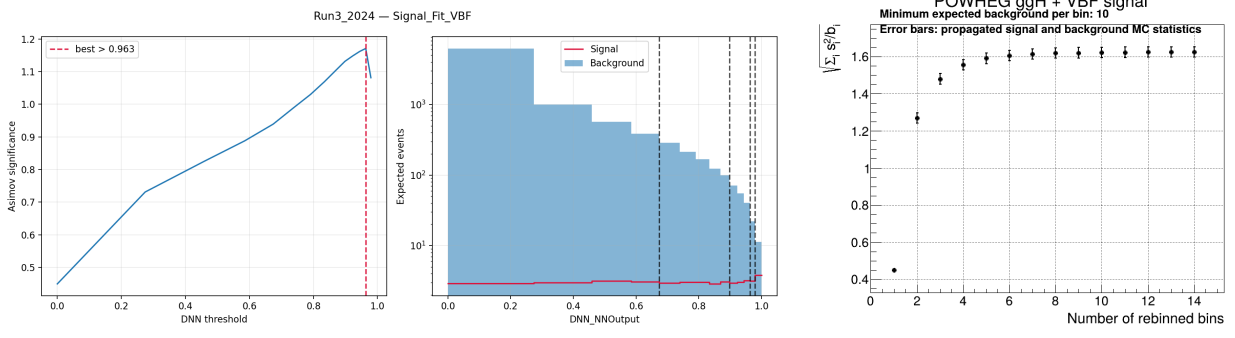


Figure 2: Left: Asimov significance against the DNN threshold and optimised edges, 2024. Right: $\sqrt{\sum_i s_i^2/b_i}$ against the number of bins, 2024, POWHEG signal. Links: [binning](#) and [sensitivity](#).

2.9 Results: data/MC

Figure 3 shows the score in the HSB of the VBF category, with the Hard/PU composition of every process. The SR is blinded above DNN = 0.8. At high score the forward-forward tag-jet configuration (FF) has the worst data/MC: +18% in 2024 and +30% in 2025 for DNN > 0.94. All eras and regions are in [dnn/data_mc](#), and the systematic shapes in [dnn/systematics](#).

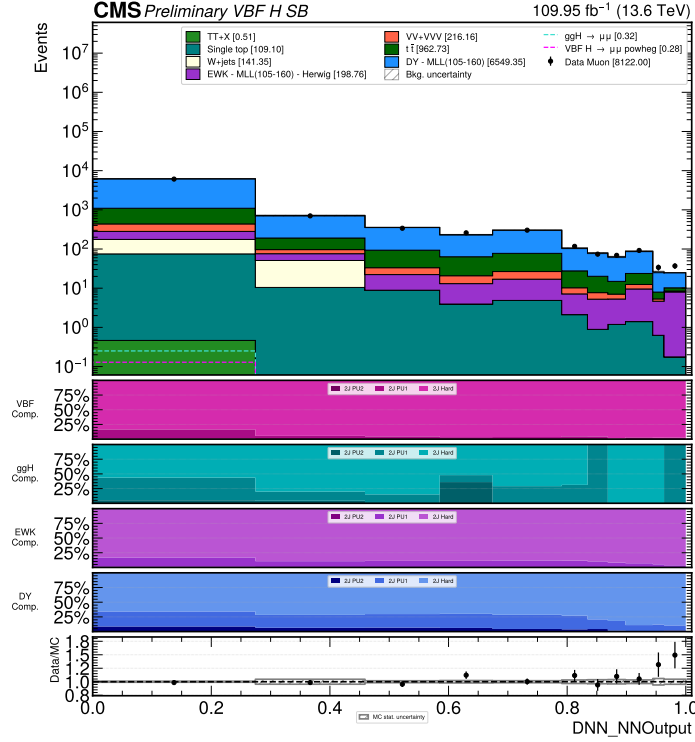


Figure 3: DNN_NNOutput in the HSB of the VBF category, 2024, with the Hard/PU composition of VBF H, ggH, EWK Z and DY.

2.10 Combine fit set-up

- **Channels:** Signal_Fit_VBF and H_sideband_VBF, one pair per era. Each era has 14 bins, and the SR is split at DNN = 0.8.
- **Signal:** VBF H + ggH with a single POI r and `--expectSignal1`.
- **Systematics:**
 - shape: JER in 6 η/p_T bins, JES as 11 regrouped sources (or the total), muon efficiencies, muon scale and resolution (ScaRe), PDF, pile-up, QCD scales, and the parton shower for

EWK Z and VBF H;

- normalisation: luminosity and cross-section $\ln N$, built per source and weighted by the dataset yields;
- MC statistics through `autoMCStats` (Barlow–Beeston lite).

- **DY models** (`--dy-model`):

- `1nn`: Hard, PU1 and PU2 with an $\ln N$ from the error of the 0/1/2-jet fit;
- `rateparam-pu`: PU1+PU2 free, Hard $\ln N$;
- `rateparam-both`: Hard and PU free;
- `inclusive`: one free DY template.

- **Asimov datasets:**

- *pre-fit*: nuisances at their nominal values;
- *toysFreq* (the reference): the nuisances are first fitted to the data with $r = 1$, with the SR above $DNN = 0.8$ masked, and the Asimov is then generated from that point.

The old one-line `--toysFreq` also fitted the high-score SR data and is no longer used.

- **Stat-only:** `--freezeParametersallConstrainedNuisances`; free rateParams stay free.

- **Tools:** `combine/build_signal_extended_datacards.py`, `combine/fit_suite.py` and `combine/run_fit_matrix.sh` (every era and all eras).

2.11 Results: expected significance

These results are the fit matrix of 4 October 2026, with seven eras. They used the old `dnn_vbf_weighted` templates (see the caveats).

DY model	SR, pre-fit	SR, toysFreq	SR+HSB, pre-fit	SR+HSB, toysFreq	r , SR+HSB, toysFreq
<code>1nn</code>	1.964	1.932	2.135	2.102	$1.00^{+0.50}_{-0.48}$
<code>rateparam-pu</code>	1.933	1.874	2.090	2.019	$1.00^{+0.53}_{-0.50}$
<code>rateparam-both</code>	1.650	1.619	1.991	1.876	$1.00^{+0.55}_{-0.54}$
<code>inclusive</code>	1.844	1.744	2.029	1.879	$1.00^{+0.55}_{-0.54}$

Table 8: Expected significance (σ), full systematics, seven eras. Stat only (`1nn`, SR+HSB, toysFreq): 2.476, with $r = 1.00^{+0.43}_{-0.42}$. Median expected limits (pre-fit, 30 September): 0.918 (`1nn`), 0.938 (`rateparam-pu`), 0.988 (`rateparam-both`).

era	SR+HSB pre-fit	SR+HSB toysFreq	toysFreq stat only	SR only pre-fit / toysFreq
2022	0.326	0.323	0.365	0.325 / 0.323
2022EE	0.578	0.564	0.640	0.567 / 0.544
2023	0.406	0.409	0.488	0.405 / 0.408
2023BPix	0.323	0.318	0.363	0.320 / 0.311
2024	1.365	1.313	1.524	1.266 / 1.259
2025	1.288	1.284	1.513	1.132 / 1.124
2026	0.707	0.711	0.783	0.693 / 0.704
all	2.135	2.102	2.476	1.964 / 1.932

Table 9: Per era, `1nn`.

- **toysFreq vs pre-fit.** toysFreq is 1.5% (`1nn`) to 7.4% (`inclusive`) lower than the pre-fit Asimov. The data pull the nuisances in the direction that reduces the sensitivity, more where DY is freer.

- **HSB.** It adds 8.8% (`1nn`) and 15.9% (`rateparam-both`).
- **Leading impacts** (`1nn`, `SR+HSB`, `toysFreq`):
 - `QCD_fac_scale_v` 0.127, mostly from EWK Z at high score;
 - JES Absolute 0.058 and RelativeBal 0.054, which are not reliable with the old templates;
 - MC statistics of the last SR bins 0.050;
 - `JEReta3pt02025` 0.048.
- **Statistics dominates.** Removing all constrained nuisances only reaches 2.51.
- **Pulls** of the `toysFreq` data fit: none beyond 1.8σ .
- **Six eras.** Until 30 September, without 2026: 2.006 pre-fit (`1nn`, `SR+HSB`). The 2.23 of 2 September was inflated by a missing VBF-filtered DY sample in 2024–25.

EWK Z normalisation. At high score EWK Z is 19–36% of the background and about 1.6 times the signal. In the cards it is constrained only by the $\pm 0.5\%$ generator integration error.

EWK Z normalisation	$\pm 0.5\%$	lnN 10%	lnN 30%	free
expected Z	2.14	2.07	1.86	1.63 (−23%)

Table 10: Dependence of the DNN fit on the EWK Z normalisation (`docs/ewkz_fit_motivation.md`, `results_fit_matrix/dnn_ewk_norm_test`).

The DNN separates H from the background, but it does not separate EWK Z from DY. This is the motivation for the VBF selector (§3).

2.12 Studies built on the DNN

VBF split by tag-jet η (CC/CF/FF).

- **Cards:** `studies/sensitivity_levers/central_cards.py`.
 - Central templates of `dnn_vbf_eta_regions`.
 - DY reweighted but not split into Hard/PU.
 - Only data statistics, MC statistics and luminosity.
- **Result:** only the ratio `split/incl` is meaningful.

eras	Asimov	incl	CC/CF/FF	gain
6	one-line <code>--toysFreq</code> (SR in the data fit)	2.255	2.429	+7.7% (artefact)
6	pre-fit	2.236	2.278	+1.9%
6	pre-fit, free DY per era and η category	2.225	2.267	+1.9%
7	pre-fit	2.350	2.393	+1.8%
7	toysFreq, whole SR masked	2.366	2.410	+1.8%
2026	pre-fit	0.726	0.732	+0.8%

Table 11: Expected significance, inclusive VBF against CC/CF/FF, no shape systematics.

- The +7.7% quoted until 30 September came from fitting the MC-statistics parameters to the high-score FF data, which have an excess.
- The statistics-only binned estimate on 2022–2025 was +4.3%.

Definition of the VBF jet pair. Three definitions were compared: $\max m_{jj}$ (production), hardest, and two leading.

all eras, 308.8 fb ⁻¹	$\max m_{jj}$	hardest	two leading
correct pair in VBF signal	0.82	0.87	0.93
VBF signal pair with ≥ 1 pile-up jet	0.086	0.058	0.034
$S/(S+B)$, SR, DNN > 0.8	0.043	0.044	0.051
signal kept	100%	100%	79%
Combine, production DNN, 1nn	1.976	2.021	2.064
small DNN per definition (2024), 1nn	1.440 ± 0.018	1.373 ± 0.062	1.414 ± 0.047

Table 12: VBF pair definitions. The small DNN is an MLP 64–32 on the 31 production inputs, retrained for each definition, with 2 folds on the event parity and three seeds.

With a dedicated network the three definitions have the same sensitivity within the spread between seeds. The production pair is kept.

Horn veto in 2025–26. The veto is needed in both eras: without it, the $2.5 < |\eta| < 3.0$ window is the worst-modelled part of the jet η spectrum. It moves 11% of VBF H from VBF to ggF and costs at most 2% in the VBF category.

2.13 Retraining on seven eras

- **Status.** The config is ready in `studies/dnn/retraining/` (30 September) but the network has not been trained.
- **Set-up:**
 - the same 32 features and architecture;
 - `era_code = 6` for 2026;
 - 4 folds on `FullEventId % 4`;
 - 200 epochs with early stopping (patience 10);
 - output in `/eos/user/v/vdamante/H_mumu/skim_v4/dnn_retraining/run3_all_eras_20260930`.
- **Comparisons:**
 1. the same features, to isolate the effect of 2026 and of the training length;
 2. without `m_mumu`, to test mass sculpting.
- **Replacement rule.** The production payload is replaced only after a comparison with the same campaign, full systematics and the same card.

3 The VBF selector (VBFSe1_NNOutput)

3.1 Purpose and signal definition

- **What it is.** A binary classifier of the VBF topology.
 - Signal: VBF H (VBFHto2Mu) + EWK Zjj (EWK).
 - Background: DY, ggH, $t\bar{t}$, VV.
- **No dimuon mass.** The mass is not an input, so the score has the same meaning in the sidebands and in the SR.

- **What “signal” means in each region.** In the sidebands the VBF signal is almost entirely EWK Z: in the 2024 HSB there are 0.6 VBF H + ggH events against 199 EWK Z. In the SR, VBF H is a small fraction of the VBF signal.
- **Measurement.** Fitting the score in the Z and H sidebands therefore measures the EWK Z signal strength r_{EWK} , and DY Hard/PU at the same time. EWK Z behaves like a standard candle of the VBF topology, and its normalisation is the leading nuisance of the $H \rightarrow \mu\mu$ fit.

Uses.

1. EWK Z normalisation in the sidebands, propagated to the $H \rightarrow \mu\mu$ card as a constraint on EWK Z (today it is constrained only at $\pm 0.5\%$).
2. A *selector-defined VBF category* (score > cut) replacing the cut-based one. This is still open.
3. Validation, era by era, of the modelling of the tag-jet kinematics and of pile-up jets: a data/MC test of the VBF topology independent of the dimuon mass.

The current $H \rightarrow \mu\mu$ pipeline still uses the cut-based VBF category.

3.2 Training sample

selection	baseline && leadingjet_pt > 0 && subleadingjet_pt > 0, i.e. baseline_ge2J
models	sidebands_only: ZSB + HSB, background cap 3×10^6 inclusive_regions: ZSB + HSB + SR, cap 1.5×10^6 ; deployed since 29 September
signal samples	VBF H powheg; EWK Zjj: inclusive (EWK_2L2J_madgraph_herwig) in the ZSB, $m_{\ell\ell}$ 105–160 (Herwig) in the HSB and SR
background samples	DY M-50 (ZSB), DY 105–160 (+ _Fil_VBF in 2024–26) in the HSB and SR, ggH, $t\bar{t}$ (2L2 ν , 4Q, L ν 2Q), WW/WZ/ZZ. No single top, W or VVV
eras	all seven (2022–2026), skim_v4; no era input
event weight	weight_Central with its sign (abs_train_weight=false)
class balance	signal sum = background sum (equalize_for_training)
cap	per background (process, region, era); random subsample (seed 12345) reweighted by N/N_{cap} , so yields are unchanged; VBF H and EWK Z never subsampled
folds	$k = 4$, test fold FullEventId % 4; the rest split 68/32 train/validation, stratified by process

Table 13: Training sample of the VBF selector (studies/vbf_selector/).

Pipeline.

1. `make_training_tuples.py`: flat tuples per era and dataset from the skims, with sample routing to the regions.
2. `build_train_sample.py`: capping and the single pickle (`train_sample_zhsr.pkl`, 8.1 GB).
3. `model_train/`: training on GPU (`main_condor.py`), one job per fold, and ONNX export.
4. `evaluate.py`: AUC per era and region, plus the cut scan.
5. `shap_per_era.py`: SHAP values.

3.3 Input features (24)

- `pt_mumu`, the only dimuon quantity;
- `N_SelectedJets`;
- for each of the four leading jets (`leadingjet`, `subleadingjet`, `thirdjet`, `fourthjet`): p_T , η , ϕ , `btagPNetQvG` (quark/gluon), `btagPNetCvL` (charm/light). Missing jets get a default value;

- `SoftJetCleanedActivity_N` and `SoftJetCleanedActivity_ptSum`.

The inputs and the network come from the VBFNet of PR #21 (A. Yeagle). What changed is the training region, the training length (one epoch in the PR) and a dropout bug fix: in the PR the dropout stayed active in evaluation mode.

3.4 Architecture and training procedure

layers	24 → 100 → 100 → 100 → 100 → 50 → 50 → 1
hidden block	Linear → BatchNorm1d → ReLU → Dropout($p = 0.10$)
output	sigmoid; the network runs in float64
loss	per-event BCE × training weight, averaged
optimiser	Adam, learning rate 5×10^{-3} , weight decay 0
batch size	10 000, shuffled
duration	early stopping on the validation loss (patience 10), best state restored
standardisation	weighted mean and std. of the training split, embedded in ONNX

Table 14: VBF selector (`common/vbfssel_configs/config.toml`). The config sets a 200-epoch limit, but the early-stopping loop does not apply it.

Deployment.

- Models: `common/vbfssel_models/trained_model_{0..3}.onnx`, byte-identical to `results_zhsr/fold_k`.
- Loss curves: `/eos/user/v/vdamante/H_mumu/skim_v4/vbf_selector/results_zhsr/fold_<k>/<k>_loss.svg`.
- The folds were trained on 29 September between 13:17 and 15:08.
- In the framework, NaN or infinite inputs are set to -10^4 and a NaN score to 0.
- Fit binning: edges at 0, 0.1, ..., 0.8, 0.85, 0.9, 0.93, 0.95, 0.97, 0.98, 0.99, 1.

Histogram campaigns.

- `vbf_selector_ewkz`:
 - score and the 24 inputs, in the ZSB and HSB;
 - categories `baseline_ge2J`, `ggF_ge2J`, `VBF_ge2J`, `ggF` and `VBF`;
 - all systematics, DY/EWK jet components.
- `vbf_selector_ewkz_eta`: the same in the VBF category split `incl/CC/CF/FF`.

3.5 Results: separation power

region	inclusive_regions			sidebands_only		
	all	VBF H only	EWK Z only	all	VBF H only	EWK Z only
Z sideband	0.8995	0.8281	0.8995	0.8998	0.8184	0.8998
H sideband	0.8237	0.8309	0.8237	0.8180	0.8224	0.8180
Signal_Fit	0.8217	0.8447	0.8200	0.8157	0.8372	0.8141

Table 15: Weighted ROC AUC, all eras, each event scored by the fold that did not see it. `sidebands_only` never saw the SR; adding the SR to the training gains about 0.006.

	2022	2022EE	2023	2023BPix	2024	2025	2026
Z sideband	0.900	0.900	0.902	0.902	0.900	0.897	0.904
H sideband	0.859	0.856	0.857	0.862	0.823	0.812	0.821
Signal_Fit	0.853	0.854	0.851	0.856	0.821	0.811	0.819

Table 16: AUC per era (inclusive_regions).

The drop from about 0.855 (2022–23) to about 0.815 (2024–26) follows the change of the horn veto. In 2024–26 it covers only $2.5 \leq |\eta| < 3.0$, and 58–63% of the VBF-category DY has a pile-up tag jet, against 16–21% in 2022–23.

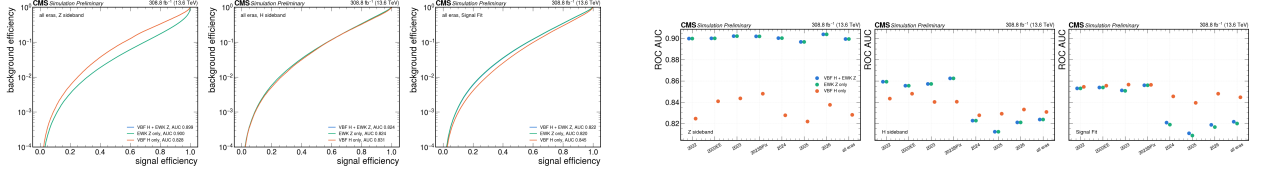


Figure 4: VBF selector (inclusive_regions): ROC curves per region, all eras (left), and AUC per era and region (right). All figures; the same set for sidebands_only.

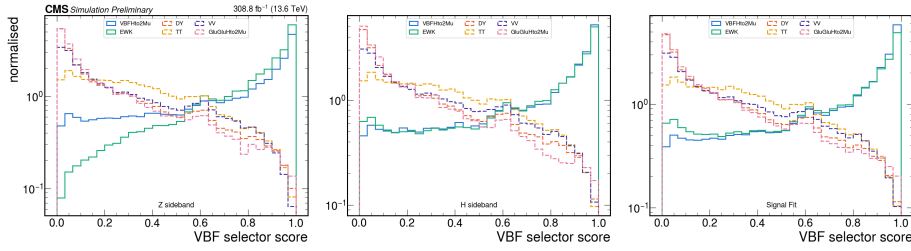


Figure 5: Score distributions of signal and background, all eras.

SHAP.

- Method: `shap.Explainer` on the fold-0 model, using events of the held-out fold, half signal and half background.
- In every era and region, and for both models, the five leading inputs are `leadingjet_pt`, `leadingjet_eta`, `subleadingjet_pt`, `subleadingjet_eta` and `pt_mumu`.
- In the SR the first one is `leadingjet_eta`.

The selector is therefore essentially a classifier of the tag-jet kinematics.

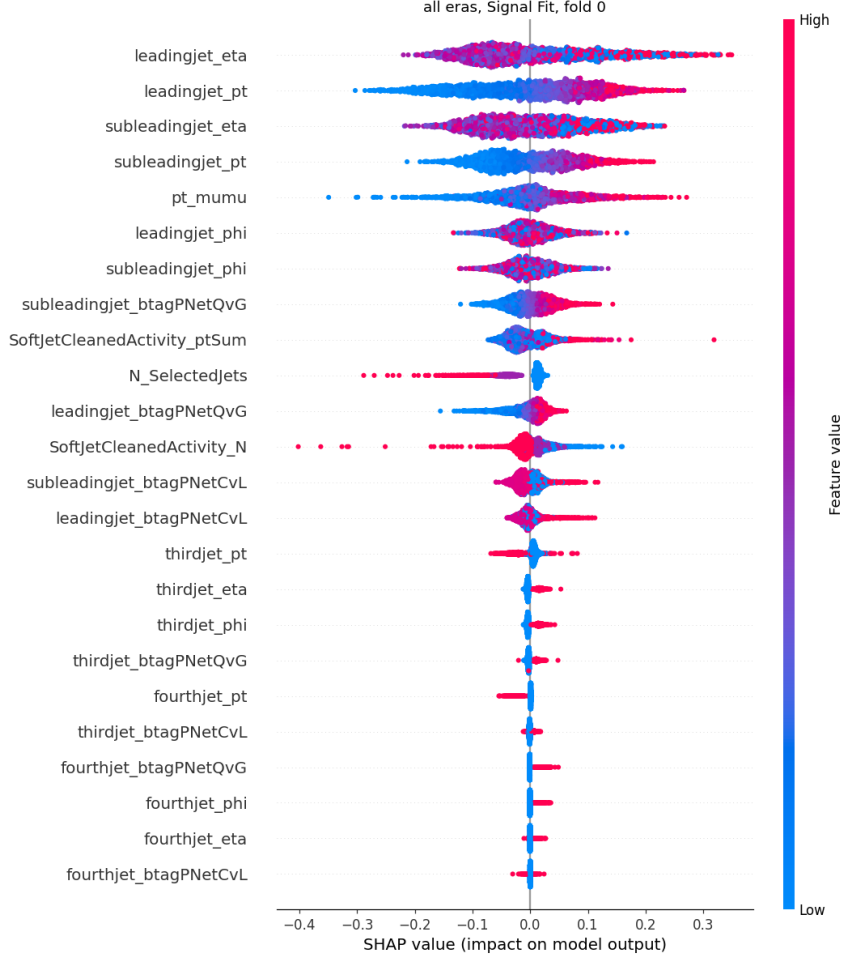


Figure 6: SHAP beeswarm in the SR, all eras. All eras and regions.

3.6 Results: selector-defined VBF category (counting)

- Scan of the score threshold in the SR, thresholds 0–0.99 in steps of 0.01 and then 0.99–1 in steps of 0.001.
- $S = \text{VBF H} + \text{ggH}$; $B = \text{DY} + \text{EWK} + t\bar{t} + \text{VV}$, with signed weights.
- The Asimov counting significances of the VBF and ggF categories are added in quadrature.
- No DNN and no systematics.

category	cut	$\epsilon_{\text{VBF H}}$	ϵ_{ggH}	ϵ_{EWK}	bkg rej.	S_{VBF}	B_{VBF}	Z_{VBF}	$Z_{\text{comb.}}$
selector	0.96	0.219	0.008	0.184	0.9944	20.4	932	0.667	1.105
cut-based	–	0.672	0.197	0.544	0.8543	113.4	24150	0.729	1.014

Table 17: All eras. Z_{ggF} : 0.882 (selector) and 0.704 (cut-based).

	2022	2022EE	2023	2023BPix	2024	2025	2026
best cut	0.97	0.97	0.97	0.95	0.97	0.96	0.96
$Z_{\text{comb.}}$, selector	0.160	0.271	0.224	0.161	0.673	0.701	0.337
$Z_{\text{comb.}}$, cut-based	0.136	0.247	0.198	0.145	0.616	0.652	0.298
gain	+18%	+10%	+13%	+11%	+9%	+8%	+13%

Table 18: Per era.

The gain comes from the ggF category, which keeps the ggH events that the cut-based VBF category takes away. The tight VBF category alone is worse than the cut-based one. This has to be repeated with the DNN fit.

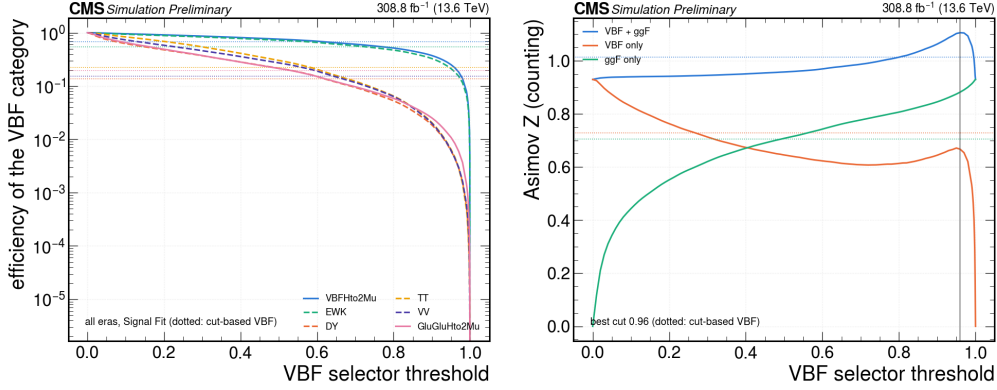


Figure 7: Cut scan on the selector score in the SR, all eras.

3.7 Results: EWK Z fit in the sidebands

Fit set-up (studies/vbf_selector/fit_ewkz_systematics.py):

- one channel per era and sideband (`zsb_<era>` and `hsb_<era>`), using the score in `baseline_ge2J`;
- POI r_{EWK} ;
- processes EWK, VBFH, ggH, TT, ST, TW, VV, VVV, TTX, VH, ttH and DY;
- lnN on luminosity and on the cross sections per source;
- autoMCStats.

DY models:

- `hard-pu` (main): DY Hard and DY PU, each with a free rateParam per channel in $[0.05, 10]$;
- `hard-pu-lnn`: as `hard-pu` with lnN constraints;
- `single`: one free DY;
- `hard-pu1-pu2`: three components, diagnostic only.

Quick fit (29 September, training tuples, no systematics, one free DY per channel): ZSB 0.918 ± 0.038 , HSB 2.08 ± 0.15 , combined 0.997 ± 0.036 . The eras range from 0.43 to 1.52. This fit is superseded.

Campaign fit (30 September, Central templates, no shape systematics):

era	Z sideband	H sideband	Z + H	obs. / exp. significance
all	1.020	1.699	1.051 $^{+0.033}_{-0.032}$	45.5 / 41.9
2022	0.596	1.626	0.616 ± 0.124	5.1 / 8.1
2022EE	0.836	1.313	0.846 ± 0.069	12.7 / 15.2
2023	0.673	0.443	0.669 ± 0.097	6.9 / 10.2
2023BPix	1.023	1.057	1.024 ± 0.140	7.4 / 7.2
2024	1.241	1.775	1.268 ± 0.055	32.5 / 25.9
2025	1.155	2.055	$1.230^{+0.086}_{-0.082}$	26.6 / 23.4
2026	1.065	1.405	$1.094^{+0.136}_{-0.130}$	8.8 / 9.3

Table 19: EWK Z signal strength on data, hard-pu (combine/results_vbf_selector_ewkz_syst/no_shape_systematics/).

- Combined fit: statistical error ± 0.013 ; expected significance 44.3 with toysFreq.
- Separate sidebands: ZSB 44.0/41.1 and HSB 12.6/8.2 (observed/expected).
- Other DY models, all eras, Z+H:
 - hard-pu-1nn: 1.118 ± 0.028 ;
 - single: 0.804 ± 0.032 , which fails in 2025 (0.28).

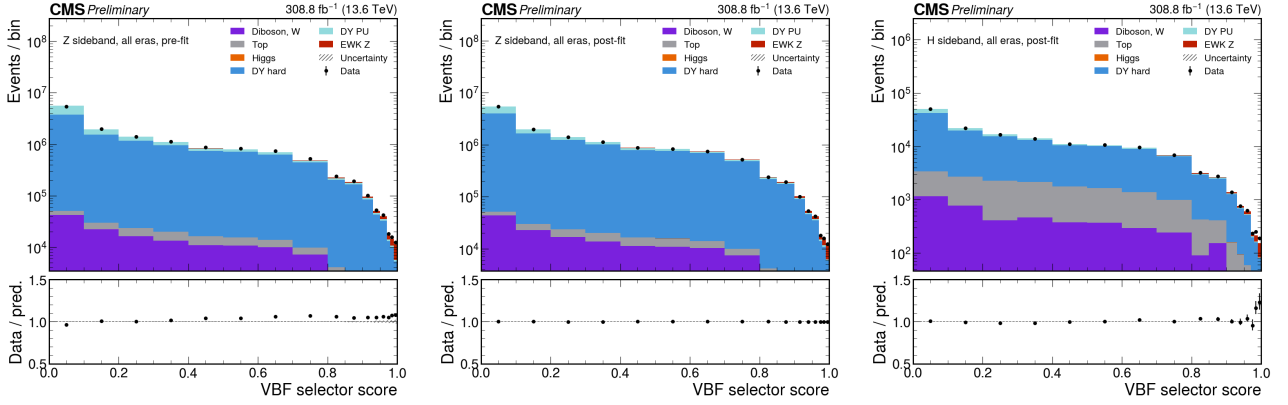


Figure 8: EWK Z fit, hard-pu: pre-fit in the Z sideband, and post-fit in the Z and H sidebands. All models and eras.

Not yet a measurement.

- The eras range from 0.62 to 1.27, and the two sidebands disagree (Z 1.02, H 1.70).
- Pre-fit data/DY in the ZSB rises from 0.96 at low score to about 1.07 above 0.7. Era by era, the ZSB data/MC goes from 1.05 down to 0.90 in 2022–22EE, is flat in 2023–23BPix, is +5% in 2024 and +10–20% in 2025–26.
- The Hard/PU split of DY reduces the sideband disagreement from 2.1/0.92 to 1.7/1.02, but the DY shape is still mismodelled. In the data fit the ZSB MC-statistics parameters are pulled by up to 7σ (2025).
- Shape systematics are needed before r_{EWK} can be used.

4 Comparison of the two networks

	analysis DNN	VBF selector
question	is it $H \rightarrow \mu\mu$?	is it a VBF topology?
signal	VBF H + ggH	VBF H + EWK Zjj
training region	cut-based VBF category, SR	<code>baseline_ge2J</code> , ZSB + HSB + SR
dimuon mass	input (<code>m_mumu</code>)	not used
inputs	32: muons, dimuon, angles, tag jets, soft activity, era	24: four leading jets, $p_T^{\mu\mu}$, soft activity
network	4×86 , ReLU, no regularisation	$4 \times 100 + 2 \times 50$, BatchNorm + dropout 0.1
training eras	2022–2025 (with <code>era_code</code>)	2022–2026 (no era input)
weights	absolute value	signed
typical AUC	0.89–0.91 (SR, VBF)	0.90 (ZSB), 0.82 (HSB, SR)
use	fit variable of $H \rightarrow \mu\mu$	EWK Z fit; alternative VBF category
status	production; retraining ready	payload <code>VBFSe1</code> installed; systematic fit pending

5 Fit plan: what exists and what is missing

The state below is as of 7 October 2026, 15:00.

- **Order.** Everything is done first for the inclusive VBF category. Only after the inclusive chain is complete and validated is the CC/CF/FF split run.
- **Templates.** The DNN templates with full systematics are `dnn_vbf_eta_regions_syst` (`incl` sub-directory for the inclusive fits; CC, CF and FF for the split). They replace `dnn_vbf_weighted`.
- **Automation.** The chain `combine/results_fit_matrix/logs/chain_20261007d.sh` runs the remaining steps: rebuild of the hadds, merge, validation of the shifted templates, then the fits only if no critical flag.

5.1 Templates

campaign	content	state	notes
dnn_vbf_weighted	DNN, VBF, all syst.	done, invalid	score not re-evaluated for JES/JER/muon scale; to be deleted
dnn_vbf_eta_regions	DNN, incl/CC/CF/FF, Central	done	used for the CC/CF/FF study without systematics
dnn_vbf_eta_regions_syst	DNN, incl/CC/CF/FF, all syst., Hard/PU	in progress	all seven eras merged on 7 October; broken hadds found (ScaRe DY M-50 2024-25, QCD scales DY 2025, Muon 2022/2025, JER up=down 2022-22EE), being rebuilt and validated
dnn_vbf	DNN, VBF, five regions, all syst.	in progress	for plots; jobs running
vbf_selector_ewkz	selector + 24 inputs, ZSB/HSB, all syst.	in progress	merge-syst only for 2022; VBF H powheg+Herwig 2024-26 and a few datasets added; 2024 preliminary merge in use
vbf_selector_ewkz_eta	selector, VBF incl/CC/CF/FF, all syst.	in progress	histogram jobs running; not merged
vbf_eta_regions_syst	all variables, incl/CC/CF/FF, all syst.	in progress	data/MC with systematic bands per η category

Table 20: Template campaigns.

Known template issues, which are expected and not bugs:

- JEReta2pt0 and JEReta3pt0 ($|\eta| > 2.5$, $p_T < 50$ GeV) have no effect, because those jets are removed by the horn veto.
- The QCD scales of EWK Z are flat, because it is an LO electroweak process.
- 2026 has no muon scale factors and therefore no muon-efficiency uncertainty. A prescription is still to be decided (2025 scale factors or a flat value).
- 2026 has no luminosity nuisance of its own and uses lumi_2025.

5.2 $H \rightarrow \mu\mu$ fits with the DNN

fit	systematics	state	result / output
<i>Inclusive VBF category</i>			
D1 SR + HSB and SR only, 4 DY models, pre-fit and toysFreq, each era and all eras (fit matrix)	full, old templates	done, invalid	2.102 (1nn, toysFreq); results_fit_matrix/dnn_sr, dnn_sr_hsb
D2 as D1, 6 eras (2022-25), one-line --toysFreq	full, old templates	done, invalid	1.998; superseded
D3 SR + HSB, DY scale V shape-only	full, old templates	done, invalid	2.172 pre-fit (30 Sept.)
D4 EWK Z normalisation test ($\pm 0.5\%$, 10%, 30%, free)	full, old templates	done	2.14 $\rightarrow$ 1.63
D5 data fit of ZSB + HSB with the SR masked (post-fit sidebands)	full, old templates	done	dnn_sr_hsb_zsb

	fit	systematics	state	result / output
D6	SR + HSB incl, Central cards (stat + lumi + MC stat), pre-fit and toysFreq	none	done	2.350 / 2.366
D7	SR + HSB and SR only, 4 DY models, significance, limits, impacts, scans (<code>results_7eras_v2/: base,</code> <code>SRonly, DYscaleShapeOnly</code>)	full, new templates	missing	runs in the chain after validation
D8	fit matrix v2: each era and all eras, pre-fit and toysFreq, impacts everywhere, post-fit HSB (<code>results_fit_matrix_v2</code>)	full, new templates	missing	runs in the chain after D7
D9	stat-only and per-group breakdown on D7/D8	–	missing	from <code>fit_suite.py</code>
<i>VBF split CC/CF/FF (only after the inclusive fits are validated)</i>				
D10	CC/CF/FF, Central cards, 6 and 7 eras, pre-fit and toysFreq (SR masked)	none	done	+1.8% (2.366 → 2.410)
D11	CC/CF/FF with free DY per era and η category	none	done	+1.9%
D12	CC/CF/FF, 4 DY models, Hard/PU split, all eras (<code>results_7eras_v2/CCCCFF</code>)	full, new templates	missing	in the chain; checks that the +1.8% survives systematics
D13	CC/CF/FF fit matrix per era, impacts	full, new templates	missing	not yet scheduled
D14	FF data/MC at high score: dedicated DY treatment or FF nuisance	–	missing	needed if FF is kept
<i>Later</i>				
D15	$H \rightarrow \mu\mu$ card with the EWK Z constraint from the selector fit (V3)	full	missing	after V3
D16	retrained DNN (7 eras; with and without <code>m_mumu</code>), same campaign and card as D7	full	missing	training not started
D17	DNN fit in the selector-defined VBF category	full	missing	needs a new campaign

Table 21: Fits with the analysis DNN. “Old templates” = `dnn_vbf_weighted` (jet/muon systematics invalid); “new templates” = `dnn_vbf_eta_regions_syst`.

5.3 EWK Z fits with the VBF selector

	fit	systematics	state	result / output
<i>baseline_ge2J, no η split</i>				
V0	quick fit on the training tuples, one DY per channel	lumi + flat lnN	done	0.997; superseded (<code>combine/archive/</code>)

	fit	systematics	state	result / output
V1	campaign templates, Central, ZSB, HSB and Z+H, each era and all eras, DY models single , hard-pu , hard-pu-lnn , hard-pu1-pu2	lumi + xsec + MC stat	done	1.051 ± 0.033 (hard-pu)
V2	fit matrix of V1: pre-fit and toysFreq Asimov, data, pulls, impacts	as V1	done	expected 41.9 / 44.3, observed 45.5
V2b	2024 only, preliminary merge, all shape systematics, single/hard-pu/hard-pu-lnn	full	in progress	combine/results_vbf_selector_ewkz_2024_prelim; a check before the full set
V3	all eras, all shape systematics, three DY models (results_vbf_selector_ewkz_syst/systematics)	full	missing	in the chain; needs merge-syst for every era
V4	fit matrix of V3 (results_fit_matrix_v2/vbf sel_systematics)	full	missing	in the chain after V3
V5	pre-fit data/MC of the score and of the 24 inputs with the systematic band	full	missing	after the merge
<i>VBF category split CC/CF/FF (only after V3–V5)</i>				
V6	selector templates in the VBF category per η class (vbf_selector_ewkz_eta)	full	in progress	histograms only
V7	EWK Z fit per η class (r common or per class), with the DY Hard/PU split	full	missing	card builder to be extended; tests whether the FF excess is EWK Z or DY
V8	performance and data/MC of the selector per η class	–	missing	
<i>Selector as a category</i>				
V9	selector-defined VBF category: cut optimisation with the DNN fit, full systematics	full	missing	counting gives +9%, from the ggF side

Table 22: Fits with the VBF selector.

5.4 Order of execution

1. **Templates, inclusive.** Rebuild the broken hadds of `dnn_vbf_eta_regions_syst` and `vbf_selector_ewkz`, merge-syst for every era, and `tools/validate_variations.py` with no critical flag.
2. **DNN, inclusive, full systematics:** D7 and D8 (with D9). This gives the new reference significance, which replaces 2.102.
3. **EWK Z, inclusive, full systematics:** V3, V4 and V5. Check whether the two sidebands and the eras become compatible once the shape systematics are included.
4. **Checkpoint:** report the inclusive results before any split.

5. **CC/CF/FF:** D12 and D13 for the DNN, V6–V8 for the selector, and D14 if FF is kept.
6. **Combination:** D15 (EWK Z constraint in the $H \rightarrow \mu\mu$ card). Then D16 (retrained DNN) and D17/V9 (selector category).

5.5 Open decisions

- 2026 muon scale factors (2025 values or a flat uncertainty) and the 2026 luminosity nuisance.
- Which DY model is the reference for the $H \rightarrow \mu\mu$ fit (1nn today) and for the EWK Z fit (hard-pu).
- Whether to keep FF, given its data/MC at high DNN score.
- Whether to remove `m_mumu` from the DNN.

References in the repository

- DNN:
 - `common/updated_DNN_configs/config.toml`;
 - `common/dnn_application.py`;
 - `studies/dnn/README.md`, `studies/dnn/retraining/README.md`;
 - `config/campaigns/dnn_vbf*.sh`.
- Fits:
 - `combine/README.md`, `combine/RESULTS.md`, `combine/COMPARISON_20261006.md`;
 - `combine/results_fit_matrix/SUMMARY.md`, `combine/results_fit_matrix/VALIDATION_20261007.md`;
 - `docs/status.md`.
- VBF selector:
 - `studies/vbf_selector/{README,RECAP,RESULTS,STATUS}.md`;
 - `common/vbfsel_configs/config.toml`;
 - `config/campaigns/vbf_selector_ewkz*.sh`;
 - `docs/ewkz_fit_motivation.md`.
- Plots:
 - https://vdamante.web.cern.ch/H_mumu/skim_v4/studies/dnn/, https://vdamante.web.cern.ch/H_mumu/skim_v4/dnn/;
 - https://vdamante.web.cern.ch/H_mumu/skim_v4/studies/vbf_selector/;
 - https://vdamante.web.cern.ch/H_mumu/skim_v4/fits/.